

Exploratory Data Analysis (EDA) Report

Titanic Dataset

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Project Objective:

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset using Pandas, Matplotlib and Seaborn to understand the data, identify trends and generate meaningful insights.

Dataset Information Dataset:

Titanic Dataset Source: Kaggle

Rows: 891 Columns: 12 File

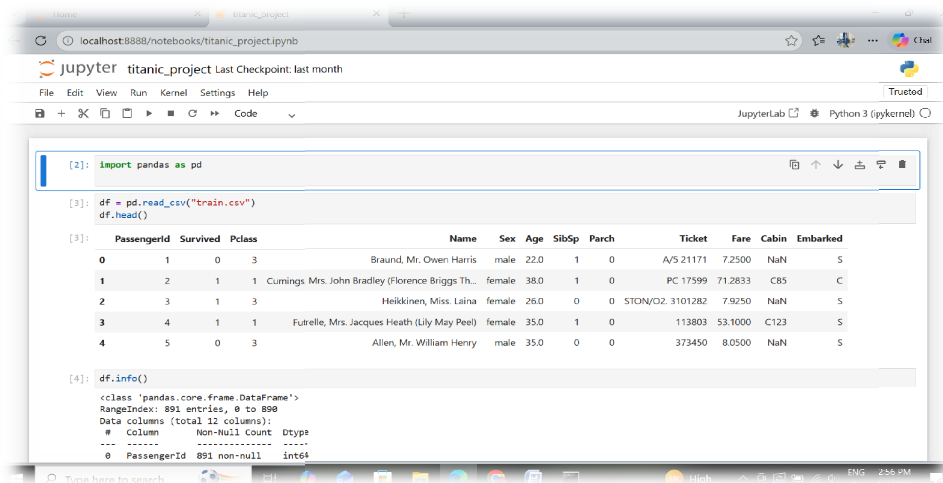
Type: CSV

Data Exploration

Functions Used:

head(), info(), describe(), shape, columns, isnull().sum(), value_counts().

Insert screenshots of these outputs in this section.



```
[2]: import pandas as pd

[3]: df = pd.read_csv("train.csv")
df.head()

[3]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
[4]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype  
---  --
 0   PassengerId         891 non-null    int64   
```

df.head()

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column             Non-Null Count  Dtype
---  -
0   PassengerId        891 non-null    int64
1   Survived            891 non-null    int64
2   Pclass              891 non-null    int64
3   Name                891 non-null    object
4   Sex                 891 non-null    object
5   Age                 714 non-null    float64
6   SibSp               891 non-null    int64
7   Parch              891 non-null    int64
8   Ticket              891 non-null    object
9   Fare                891 non-null    float64
10  Cabin               204 non-null    object
11  Embarked            889 non-null    object
dtypes: float64(2), int64(5), object(5)
```

```
df.shape
```

```
(891, 12)
```

```
df.columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

```
df.isnull().sum()
```

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
```

Data Visualization:

Include screenshots of Histogram, Countplot, Gender vs Survival, Passenger Class vs Survival, Boxplot, Scatterplot and Correlation Heatmap.

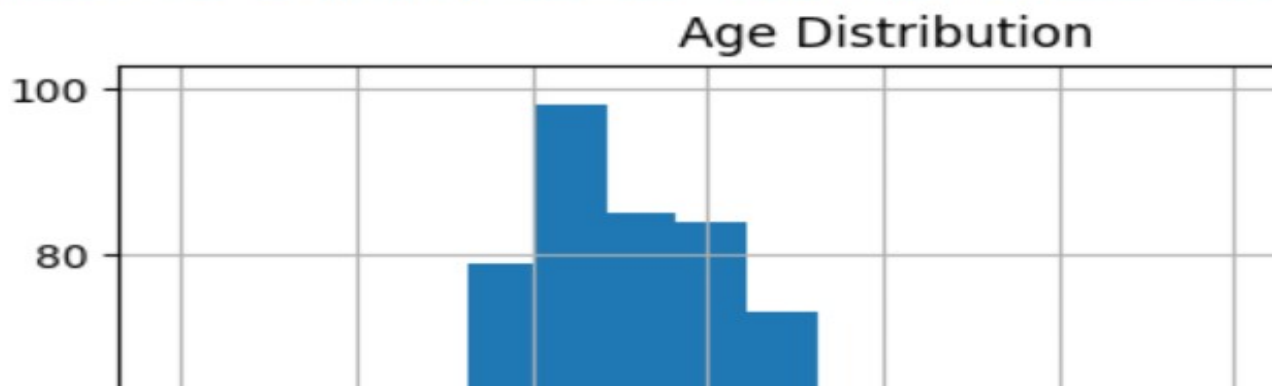
```
import matplotlib.pyplot as plt

df["Age"].hist(bins=20)

plt.title("Age Distribution")

plt.show()
```

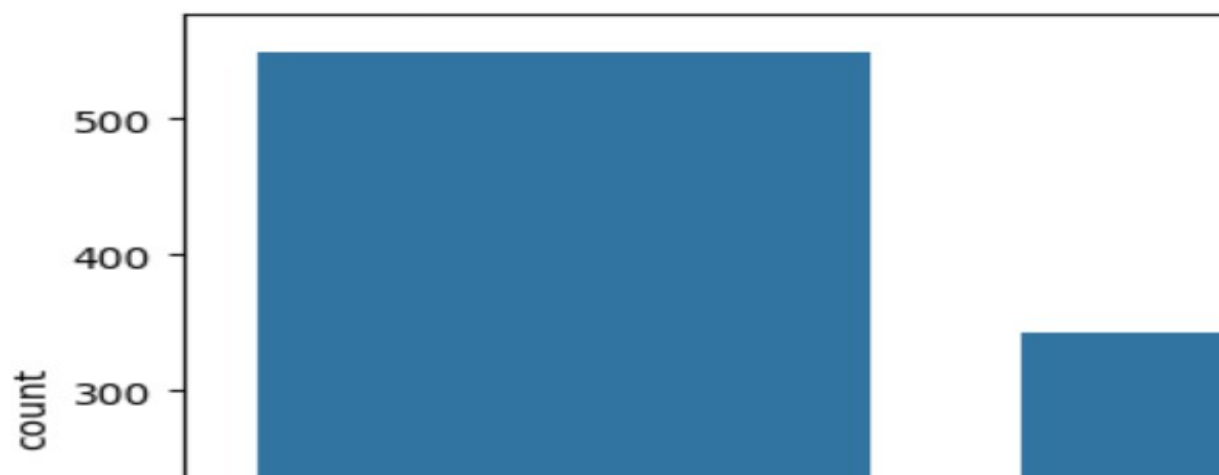
Matplotlib is building the font cache; this may take a moment



```
import seaborn as sns

sns.countplot(x="Survived", data=df)

plt.show()
```

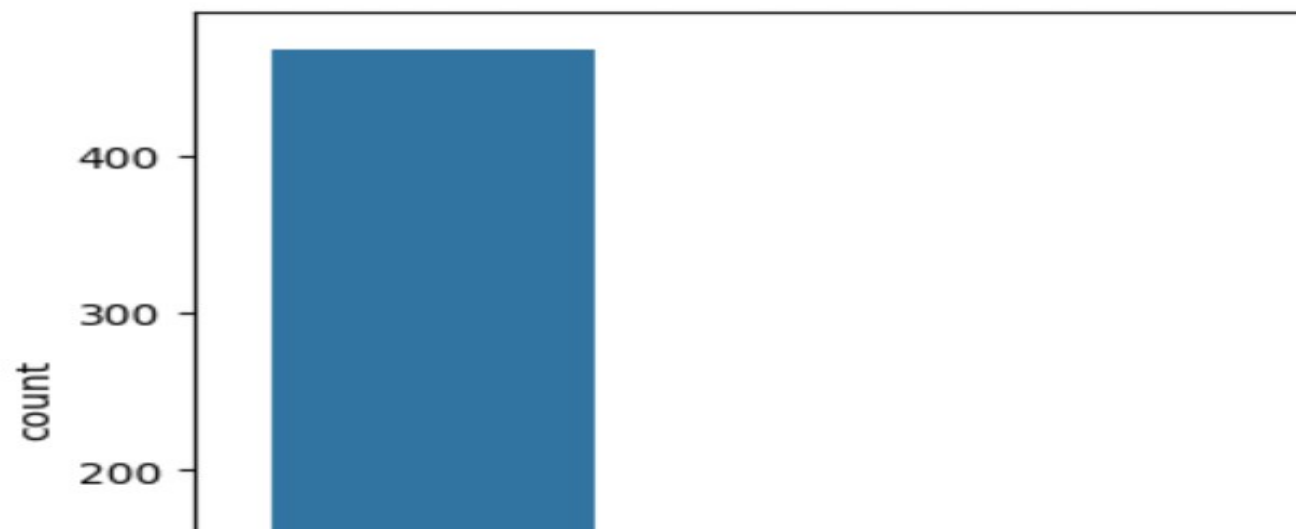


Key Findings –

Female passengers had a higher survival rate. - First-class passengers survived more frequently. - Age, Cabin and Embarked contain missing values. - Fare and Passenger Class influenced survival.

```
sns.countplot(x="Sex",hue="Survived",data=df)

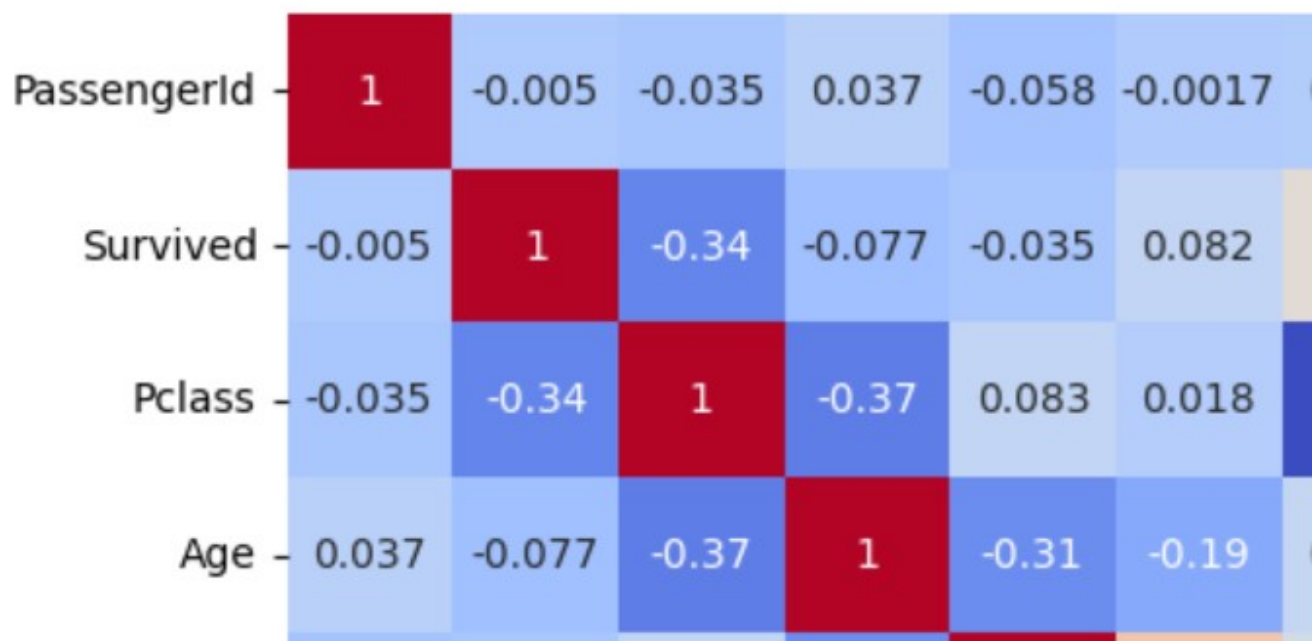
plt.show()
```



```
numeric_df = df.select_dtypes(include="number")

sns.heatmap(numeric_df.corr(),annot=True,cmap="coolwarm")

plt.show()
```



Conclusion :

The project successfully explored the Titanic dataset using EDA techniques. Visualizations and statistical summaries helped identify important trends and relationships.

References: Conclusion The project successfully explored the Titanic dataset using EDA techniques. Visualizations and statistical summaries helped identify important trends and relationships. References Kaggle Titanic Dataset Pandas Documentation Matplotlib Documentation Seaborn Documentation

Kaggle Titanic Dataset

Pandas Documentation

Matplotlib Documentation

Seaborn Documentation